Related Work Map for Angular Manifold Routing Preprint

This document summarizes the most relevant research areas adjacent to the Angular Manifold Routing architecture. The purpose is to help position the work accurately in a preprint, ensuring that prior literature is acknowledged while clearly distinguishing the proposed contribution. The key claim of the present work is that routing tokens by angular sector on a normalized hypersphere naturally produces sublinear routing occupancy, empirically approximated by: $E(K) \approx c \cdot K^\alpha$, where $\alpha < 1$ This produces automatic sparsity and hardware locality benefits without relying on learned gating networks. The literature below represents the closest conceptual neighbors to this idea.

Hyperbolic / Non■Euclidean Representation Learning

This body of work studies neural networks in curved spaces, especially hyperbolic geometry. Hyperbolic embeddings are particularly useful for representing hierarchical structures such as taxonomies, graphs, and language relationships. Closest overlap: recognition that geometry of representation space affects model efficiency. Key difference: these works focus on representation structure rather than routing architectures. They do not introduce deterministic angular routing partitions or measure routing■capacity scaling.

Standard Mixture■of■Experts Routing

Modern mixture■of■experts architectures route tokens to experts using learned gating networks. A router network computes scores for each expert and selects top■k experts per token. Closest overlap: conditional compute and sparse routing. Key difference: routing decisions are learned and optimized during training rather than emerging from geometric structure. These methods typically assume near■uniform expert usage and introduce load■balancing losses to enforce it.

Hyperspherical MoE Research

Some research has explored performing routing computations on normalized hyperspherical embeddings. These approaches typically normalize token embeddings and expert vectors, then compute routing scores using cosine similarity. Closest overlap: use of hyperspherical geometry in routing computations. Key difference: routing still occurs through learned gating scores. The geometry is used as a scoring space rather than as a deterministic sector■partition routing mechanism.

Spherical Routing Variants

Recent work has proposed spherical routing mechanisms in specific domains such as image restoration. These systems use similarity■based expert activation within spherical embedding spaces. Closest overlap: explicit spherical routing language. Key difference: these systems still rely on learned expert activation mechanisms rather than fixed angular partitioning. They do not analyze routing occupancy scaling or hardware locality benefits.

Geometry■Aware Aggregation

Some research focuses on preserving geometric relationships when aggregating outputs from multiple experts. These approaches may decompose vectors into radial and angular components on a sphere or manifold. Closest overlap: explicit angular/radial geometry modeling. Key difference: the geometry affects aggregation of expert outputs rather than the routing mechanism that determines which experts are activated.

Latent■Space Routing Improvements

Recent work proposes improving routing stability by learning low■rank or controlled latent routing spaces. These techniques aim to improve specialization and prevent instability in gating networks. Closest overlap: the idea that routing quality depends on geometric structure of embeddings. Key difference: routing remains a learned function rather than a deterministic geometric partition.

Area of Prior Work	Main Idea	How It Differs From Angular Manifold Routing
Hyperbolic Representation Learning	Uses curved spaces to represent hierarchical data efficiently	Focuses on representation geometry rather than routi
Standard MoE Routing	Uses learned gating networks to select experts	Routing learned by neural network rather than geome
Hyperspherical MoE Methods	Uses normalized embeddings and cosine similarity for routing scores	Geometry used for scoring rather than deterministic a
Spherical Routing Variants	Applies spherical similarity routing in domain■specific tasks	Still uses learned activation logic rather than sector di
Geometry■Aware Aggregation	Preserves angular and radial structure when combining expert outputs	Focuses on aggregation rather than routing structure
Latent Routing Space Methods	Learns improved routing geometry in latent space	Routing remains learned rather than deterministic

Positioning Statement for the Preprint A defensible description of novelty would be: "While prior work has explored geometric representation learning and mixture■of■experts routing, we are not aware of work combining normalized hyperspherical embeddings, deterministic angular partition routing, emergent routing sparsity via bucket compression, and an empirical routing occupancy scaling law with hardware locality implications." This phrasing acknowledges adjacent research while clearly stating the distinct contribution of the Angular Manifold Routing framework.